# Modeling Profit Give-Back Risk in Algorithmic Trading via LSTM-Based Exit Decision Systems

First Author* Second Author†

This Draft: May 4, 2026

**Abstract**

This paper studies ... We examine ... Using ... data from ... to ..., we find that ... The results suggest that ... This paper contributes to the literature by ...

**Keywords:** Stablecoins; Treasury Yields; Monetary Policy Transmission; Local Projections; Financial Markets.

**JEL Classification:** E43; E44; E52; G12; G15.

**Acknowledgments:** We thank ... for helpful comments and suggestions. All errors are our own.

*First Author: Department of Finance, XXX University. Email: first.author@email.com.
†Second Author: School of Economics, XXX University. Email: second.author@email.com.

# 1 Introduction

In quantitative trading, trend-following strategies have been widely adopted because of their structural simplicity and robustness across markets. Among them, crossover strategies based on moving averages (MA) or weighted moving averages (WMA) represent one of the most typical implementation paradigms. Such methods usually rely on moving-average crossover signals to open positions and reverse positions, and have shown a certain degree of effectiveness in capturing medium- to long-term price trends. However, although entry rules have been extensively studied and have become relatively mature, the optimization of exit timing remains one of the key factors limiting strategy performance.

Traditional trend-following strategies commonly adopt a close on reverse signal mechanism, under which a position is exited only when a trend reversal occurs and an opposite crossover signal is triggered. This mechanism is inherently subject to substantial lag. While it helps capture a complete trend interval, in the later stage of a trend or during range-bound markets, accumulated unrealized profits often cannot be locked in in a timely manner, leading to a drawdown of unrealized profits. To address this issue, existing methods typically rely on fixed take-profit/stop-loss rules or trailing stops. However, these methods depend on empirically chosen parameters, are difficult to adapt to dynamic market conditions, and lack a data-driven decision-making capability for determining whether a position should be exited early.

To systematically characterize this problem, this paper first conducts an empirical analysis of the trading paths of trend-following strategies in the cryptocurrency market and identifies a salient empirical fact that has not been sufficiently discussed: the vast majority of trades once achieved positive unrealized returns during the holding period, but only a small proportion eventually closed with positive returns. Specifically, in the dataset examined in this study, around 90% of trades were in profit at some point, whereas only about 30% ended as profitable trades. This finding suggests that the main bottleneck of trend-following strategies is not the absence of profit opportunities, but the lack of an effective mechanism for profit locking and risk control.

Based on the above observation, this paper formalizes the phenomenon as unrealized profit drawdown risk and models it as a sequential decision-making problem: during the holding period, how can one determine, based on the current market state and position-related characteristics, whether continued holding will lead to a significant future drawdown in returns, and thereby decide whether the position should be exited early? Unlike traditional approaches that focus on price prediction or trend identification, this paper emphasizes the dynamic evolution of the profit path and its associated risk structure.

To this end, this paper proposes an early-closing decision framework based on Long Short-Term Memory (LSTM). The proposed method formulates the exit problem as a sequence classification task. By taking as input time-series data containing both market features and position-context information, the model learns the probability of executing a closing operation at the current time. Furthermore, this paper designs a value-based labeling function grounded in the trade-off between future returns and drawdowns, enabling the model to directly learn the decision boundary of whether "continued holding is preferable to early exit." From a system perspective, the model can be regarded as a data-driven decision layer embedded on top of a traditional trend-following strategy, with the aim of improving the quality of exit decisions. In the empirical analysis, this paper conducts backtesting based on the cryptocurrency market. The results show that, while overall returns remain broadly stable, the proposed method significantly improves the strategy's win rate and Sharpe ratio, and effectively reduces the maximum drawdown.

# 2 Literature Review

The central issue in quantitative trading research is how to identify exploitable price regularities from historical market information and transform them into trading strategies that can be implemented on a sustainable basis. Around this issue, the related literature has broadly evolved from traditional technical-analysis rules to machine-learning-based prediction, and further to reinforcement-learning-based dynamic decision making. Early studies mainly relied on trend-following rules such as moving averages, moving-average crossovers, and momentum, generating buy and sell signals directly from price series. Subsequently, machine learning methods began to be widely used for price-direction prediction, return prediction, and trading-signal identification. In recent years, reinforcement learning has further framed trading as a sequential decision-making problem, making take-profit, stop-loss, and dynamic exit optimization increasingly important research topics. Overall, the existing literature has accumulated relatively rich findings on how to identify opportunities and how to predict prices, but there remain clear limitations in how to optimize exit timing and how to reduce the drawdown of profits.

## 2.1 Trend-following Strategies

Trend-following strategies are among the most classical research directions in quantitative trading. Their basic idea is to exploit price inertia by using technical indicators to identify trends and trade in the direction of those trends. Moving-average-based methods are the most representative strategies in this field, including the simple moving average (SMA), weighted moving average (WMA), exponential moving average (EMA), and short–long moving-average crossover rules. Hunter (1986) provided an early systematic discussion of the exponentially weighted moving average, emphasizing the importance of recent information in time-series analysis; Hansun (2013) further discussed possible improvements to moving-average methods on this basis, showing that traditional moving-average tools themselves have continued to evolve.

As research developed, scholars were no longer satisfied with treating moving-average rules as purely empirical trading tools, and began to examine their effectiveness from both theoretical and empirical perspectives. Chiarella et al. (2006) analyzed the mechanism of moving-average rules from the perspective of a dynamic market model, pointing out that the length of the moving-average window and feedback from trading behavior can significantly affect price dynamics. Ellis and Parbery (2005) compared adaptive moving-average and simple moving-average strategies, showing that dynamic adjustment of moving-average parameters can help improve strategic adaptability. Marshall et al. (2017) further compared time-series momentum with moving-average rules, noting that although both are based on the idea of trend continuation, they differ in signal construction and sources of returns.

Overall, the evolution of trend-following strategies has moved from single rules with fixed windows and fixed thresholds toward parameter adaptation and more diversified rule structures. For this reason, traditional technical analysis has not lost its value with the development of machine learning. Instead, it has become an important source of features for subsequent intelligent trading models. The study by Ayala et al. (2021), for example, shows that technical-analysis strategies are increasingly being re-coded, optimized, and integrated by machine learning methods, thereby enabling a methodological transition from rule-driven to data-driven approaches.

## 2.2 Machine Learning in Trading

After machine learning entered the field of quantitative trading, the research paradigm changed markedly. Unlike traditional technical analysis, which relies on rules predefined by researchers, machine learning emphasizes automatically learning the mapping between historical data, price changes, and trading outcomes. Early representative work such as Trippi and DeSieno (1992) had already attempted to use neural networks for stock-index futures trading, indicating that researchers had long recognized the possibility that complex nonlinear relationships behind price movements could be captured by models. Subsequently, Dempster and Jones (2001) used genetic programming to improve trading rules, while Dempster and Leemans (2006) applied reinforcement learning to automated foreign-exchange trading, both reflecting the gradual shift of quantitative trading research from manual rule design toward intelligent decision making.

In terms of method types, the application of supervised learning in quantitative trading can mainly be divided into classification and regression. Classification methods are usually used to predict price directions, trend reversals, or buy and sell signals. For example, studies such as Shen et al. (2012) and Gerlein et al. (2016) focus on directional prediction and trading classification in stock markets. Regression methods, by contrast, are more often used to predict future price levels and returns, with the emphasis placed on improving numerical forecasting accuracy and supporting subsequent trade execution. Zhong and Enke (2019) predicted the daily return direction of the stock market using hybrid machine learning algorithms, demonstrating the practical value of classification models in generating trading signals. Adegboye and Kampouridis (2021) discussed the applicability of both classification and regression models to the prediction of directional-change trends and directional reversals in the foreign-exchange market, indicating that the two are not substitutes for one another in quantitative trading, but instead serve different objectives.

In this process of development, the LSTM model became an important milestone in quantitative finance research. Since financial time series typically exhibit nonlinearity, time variation, and long-term dependence, traditional machine learning models often face insufficient memory capacity when dealing with such data. Through its gating mechanism, LSTM can better retain long-term information, and thus quickly became a core tool for price prediction and trading-signal modeling. Nelson et al. (2017) used LSTM to predict stock price directions, providing an early demonstration of the model's applicability to financial sequence forecasting. Fischer and Krauss (2018) further demonstrated the significant advantages of LSTM in financial market direction prediction. Later, Borovkova and Tsiamas (2019) extended LSTM to high-frequency stock classification and improved the model's robustness in non-stationary markets through an ensemble approach. Studies such as Mehtab et al. (2020) and Ghosh et al. (2022) further show that LSTM, as well as hybrid architectures combining LSTM with CNNs, random forests, XGBoost, and other models, has become an important technical route in quantitative trading research.

More recently, the focus of machine learning research has no longer been confined to whether prediction is accurate, but has gradually shifted toward whether models genuinely improve trading performance. In a study of the cryptocurrency market, Sebastião and Godinho (2021) emphasized that model performance differs significantly across market states, suggesting that in-sample accuracy cannot be directly translated into stable trading profits. Han et al. (2023), from the perspective of label design, linked price prediction more closely with trading-system construction. This indicates that machine learning research in quantitative trading has gradually moved from pure price prediction toward a profit-oriented and strategy-oriented practical framework.

## 2.3 Exit Strategy Optimization

Compared with the large body of research focusing on the identification of entry signals, the optimal exit strategy concerns the question of when to close a position, which has a more direct impact on the realization of trading profits. Exit decisions not only determine the final profit or loss of individual trades, but also relate to profit protection and the control of risk exposure, and are therefore of great importance in actual trading.

Early studies on exit problems mainly focused on fixed stop-loss and fixed take-profit rules. Leung and Li (2015) introduced transaction costs and stop-loss exit constraints into a mean-reversion trading framework, pointing out that the setting of exit boundaries directly affects strategy optimality. Lo and Remorov (2017) further analyzed the effectiveness of stop-loss strategies in the presence of serial correlation, regime switching, and transaction costs, showing that stop-loss rules do not unconditionally improve the risk–return profile in all environments, but depend heavily on market states and parameter settings. Vezeris et al. (2018) compared the performance of multiple take-profit and stop-loss strategies in combination with an MACD system, finding that volatility-based dynamic thresholds have stronger adaptability than fixed thresholds.

On this basis, research on trailing stop-loss rules began to attract attention. The core function of a trailing stop is not merely to limit losses; more importantly, as prices move in a favorable direction, the exit threshold is raised accordingly, thereby locking in part of the paper profit and reducing the drawdown of profits. Glynn and Iglehart (1995) provided an early theoretical discussion of the trading implications of trailing stops. Dai et al. (2021) showed from a risk-control perspective that trailing stop-loss rules can effectively reduce total risk and downside risk in declining markets. Leung and Zhang (2021) further formulated trailing stops as an optimal stopping problem with path-dependent constraints, promoting the development of exit-decision research from empirical rules toward theoretical optimization models.

In recent years, reinforcement learning has provided new methodological tools for the study of optimal exits. Unlike supervised learning, which focuses on predicting future prices, reinforcement learning is better suited to handling the sequential decision problem of what action should be taken under a given market state, and is therefore naturally applicable to position management and exit-timing optimization. Kim and Kim (2019) applied deep reinforcement learning to the optimization of trading boundaries and stop-loss boundaries in pairs trading. Tsantekidis et al. (2021) proposed a deep reinforcement learning framework based on price trailing, directly improving exit quality during the trading process through a modified reward function. The DeepTrader model of Wang et al. (2021) and the review by Sun et al. (2023) further show that reinforcement learning has become an important frontier in quantitative trading, especially in dynamic asset allocation and trade execution.

However, judging from the distribution of existing studies, reinforcement learning is still more often used for overall trading decisions, asset allocation, and direction selection, rather than for the systematic optimization of the exit timing of individual trades. Hwang et al. (2023) pointed out that stop-loss mechanisms in actual trading often change the effectiveness of the original buy and sell signals, indicating that correct prediction does not necessarily imply profitable trading, and that exit rules play a decisive role in this process. Although Sivri et al. (2023) and Shawky (2023) have begun to directly use machine learning to determine stop-loss and take-profit levels, the number of such studies remains relatively limited, suggesting that optimal exit remains an important topic in quantitative trading research that has not yet been sufficiently developed.

Existing studies have produced relatively rich findings in trend identification, price prediction, and the construction of automated trading systems, but there remain several clear shortcomings in exit optimization, especially in the control of profit drawdowns.

First, although traditional trend-following strategies are intuitive and interpretable, they are highly sensitive to parameter settings, sample periods, and market environments. Once the moving-average window and trend-identification rules are set inappropriately, strategy performance may fluctuate significantly. This suggests that traditional rules often lack robustness in markets and find it difficult to adapt continuously to different volatility states and structural changes.

Second, the overall amount of research on optimal exits remains limited, and its internal development is uneven. Existing studies pay more attention to when to buy and whether prices will rise, while the questions of when to sell and how to reduce the drawdown of profits remain insufficiently discussed. In particular, compared with research on stop-loss rules, the literature on take-profit rules, trailing take-profit mechanisms, and the joint optimization of take-profit and stop-loss rules is more limited.

Third, most existing studies treat entry, holding, and exit separately, and have not yet formed a unified optimization framework running through trend identification, signal generation, and dynamic exit. In real trading, the final performance of a trade does not depend only on whether the entry point is accurate, but also heavily on risk management during the holding period and the method of exit. If the exit mechanism is poorly designed, even strong predictive ability may be eroded by profit drawdowns, frequent stop-loss triggers, or transaction costs. Therefore, future research needs to integrate prediction models and exit decisions more closely, and incorporate profit protection, drawdown control, and risk-adjusted returns into a unified objective function.

In summary, the existing literature has provided a relatively solid research foundation for trend following, machine-learning-based prediction, and reinforcement-learning-based trading, but research on optimal exit, especially the control of profit drawdowns, remains clearly insufficient. For this reason, starting from dynamic take-profit and trailing exit mechanisms, and combining them with machine learning methods to build a unified decision-making framework that is closer to real trading environments, has not only clear theoretical significance but also strong practical value.

# 3 Empirical Analysis of Profit Give-Back

## 3.1 Definition of Key Indicators

To systematically characterize the profit evolution path of the trend-following strategy during the holding period and its drawdown features, this paper defines the following key indicators:

**Maximum Favorable Excursion (MFE)**:MFE refers to the highest unrealized profit level reached by a trade over its entire holding period, and is used to measure the maximum potential profit that the trade once achieved along its path. It is defined as follows:

$$MFE = \max_{t \in [t_{entry}, t_{exit}]} PnL(t) \tag{1}$$

where $PnL(t)$ denotes the unrealized profit or loss of the trade at time $t$.

MFE reflects the "optimal exit opportunity" provided by the market during the holding period and serves as a basic reference for analyzing profit drawdown behavior.

**Final Profit and Loss (Final PnL)**:Final PnL is defined as the realized return of the trade at the time of closing:

$$PnL_{final} = PnL(t_{exit}) \tag{2}$$

This indicator represents the final locked-in return of the strategy and is a core metric for evaluating traditional performance.

## 3.2 Statistical Results

Based on the trading data of the WMA trend-following strategy for five major cryptocurrencies (DOGE, SOL, XRP, ETH, and BTC) from January 1, 2022 to 2026, this paper conducts a statistical analysis of the profit path of each trade. The results show that, across all cryptocurrencies, the proportion of trades with a maximum favorable excursion greater than zero lies between 85% and 87%, whereas the proportion of trades that eventually realize positive returns is only approximately 33% to 36%.

This result shows a high degree of consistency across different assets, indicating that the phenomenon is stable across assets. From a statistical perspective, this means that, in the vast majority of trades, the market once provided profit opportunities, but these opportunities failed to be converted into final returns. In other words, the main issue of trend-following strategies is not whether they are able to generate profits, but whether they are able to lock in the acquired unrealized profits at an appropriate time.

Table 1: Statistical Results of Trading Paths

| Asset | Total Trades | MFE > 0 Trades | MFE > 0 Ratio | Final PnL > 0 Trades | Final PnL > 0 Ratio | Average MFE (USDT) | Average Final PnL (USDT) | Total PnL (USDT) |
|---|---|---|---|---|---|---|---|---|
| DOGE | 1575 | 1371 | 87.05 | 566 | 35.94 | 36.75 | 2.79 | 4389.06 |
| SOL | 1579 | 1371 | 86.83 | 551 | 34.90 | 38.79 | 2.60 | 4103.44 |
| XRP | 1579 | 1359 | 86.07 | 552 | 34.96 | 30.58 | 1.77 | 2789.82 |
| ETH | 1607 | 1404 | 87.37 | 541 | 33.67 | 26.04 | 1.34 | 2152.53 |
| BTC | 1627 | 1399 | 85.99 | 536 | 32.94 | 19.02 | 0.91 | 1486.19 |

## 3.3 Profit Path Analysis

To further reveal the structural characteristics of unrealized profit drawdown risk, this paper analyzes the profit path. Based on the trading data of different cryptocurrencies, this paper constructs the average profit path after normalizing the holding period. The results show a highly consistent dynamic structure: in the early stage of the holding period, profits increase monotonically over time; in the middle-to-late stage, profits reach a peak corresponding to MFE; and in the later stage, a significant drawdown occurs. This pattern indicates that the profits of trend-following strategies are not accumulated monotonically, but generally experience a dynamic process of "profit formation–profit drawdown."

By analyzing the relationship between the MFE and the final PnL of each trade, it can be observed that a large number of trades satisfy $MFE > 0$ and $PnL_{final} < 0$, that is, the trade was profitable at some stage but eventually ended with a loss. This phenomenon directly reflects the existence of unrealized profit drawdown risk, and indicates that relying solely on final profit indicators cannot reflect the true profitability of the trading process.

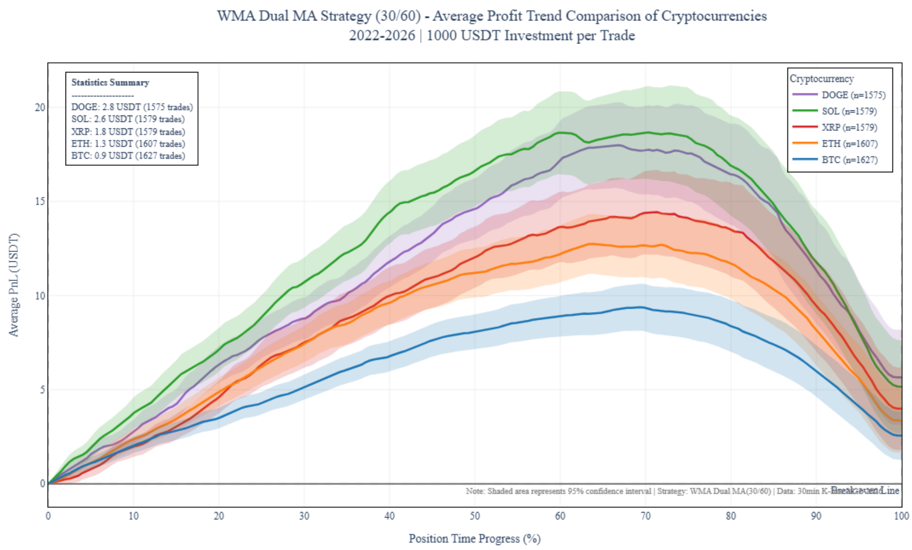


Figure 1: Average Profit Pathss Across Cryptocurrencies

Trend-following strategies are able to capture positive profit opportunities in most trades ($MFE > 0$), indicating their effectiveness in trend identification. However, only about 30% of trades are able to close with positive returns, suggesting that a large amount of unrealized profits is not effectively locked in. The profit path analysis shows that trades generally experience a dynamic process of "profit formation–profit drawdown." Therefore, it can be concluded that unrealized profit drawdown risk is a systematic and prevalent phenomenon in trend-following strategies. This finding fundamentally explains the contradiction in trend-following strategies between "high potential profits and low realized profits," and indicates that the exit problem is essentially a decision problem of "profit protection" rather than "profit acquisition." This conclusion provides a direct theoretical motivation for the subsequent proposal of an early-closing decision method based on sequence learning.

# 4 Methodology

## 4.1 Framework Overview

This paper proposes a hybrid decision-making framework that combines the signal identification capability of traditional trend-following strategies with the nonlinear risk-control capability of deep learning models. The framework consists of three key components: baseline strategy layer, LSTM decision layer, and exit execution mechanism. The baseline strategy layer uses weighted moving average (WMA) crossover signals as the logical starting point, and is responsible for capturing market trends and generating initial trading signals. The LSTM decision layer serves as an intelligent risk-management module embedded on top of the baseline strategy. By monitoring real-time market-state features and position-context information, it dynamically evaluates the risk of unrealized profit drawdown.

The core operating logic of the system is a two-layer collaborative mechanism. When the baseline strategy layer triggers an entry signal according to the moving-average crossover rule, the system enters the holding state. At this point, the LSTM decision layer is activated simultaneously. Through nonlinear mapping of the input feature sequence, it outputs in real time the conditional probability of whether an early closing operation should be executed at the current time. If the predicted probability exceeds a preset decision threshold, the system will execute an exit operation before the traditional trend-reversal signal appears, so as to lock in profits and avoid or mitigate potential large drawdown. From a system-architecture perspective, this model constitutes a data-driven decision layer superimposed on the traditional rule-based logic, thereby significantly improving the refined management capability of the strategy during the holding period.

## 4.2 Baseline Strategy

To ensure the sensitivity and representativeness of the empirical analysis, this paper selects a weighted moving average (WMA) dual moving-average system as the baseline strategy. Compared with a simple moving average (SMA), WMA assigns greater weights to more recent price observations, enabling it to respond more quickly to price changes and to exhibit greater flexibility in trend identification. The strategy logic adopted in this study follows a standard moving-average crossover rule. In terms of entry rules, the system uses the crossover logic of a 30-period fast moving average and a 60-period slow moving average. When the fast moving average crosses above the slow moving average from below, that is, when a golden cross is formed, a long-entry signal is executed. Conversely, when the fast moving average crosses below the slow moving average from above, that is, when a death cross is formed, a short-entry signal is executed. In terms of traditional exit rules, the baseline mechanism used for comparison relies only on reverse trading signals; that is, a position is closed only when an opposite signal to the current position direction appears. Although this mechanism can help capture relatively complete trend intervals, it also faces significant unrealized profit drawdown risk in the later stage of a trend, which constitutes the core motivation for introducing the LSTM decision layer in this paper.

## 4.3 Problem Formulation

In the hybrid decision-making framework proposed in this paper, the LSTM decision layer does not directly undertake the task of predicting trend direction. Instead, it is defined as a dynamic exit decision model for the holding period. In other words, the baseline strategy layer has already determined the trading direction according to the WMA crossover rule, while the core objective of the LSTM model is to judge whether the current unrealized profit faces a relatively high future drawdown risk during the holding process.

Let the market feature vector at time $t$ be denoted by $\mathbf{x}_t$. For any trading sample that is in the holding state, the model takes the feature sequence over the past $k$ time steps as input:

$$\mathbf{X}_{t-k:t} = \{\mathbf{x}_{t-k}, \mathbf{x}_{t-k+1}, \cdots, \mathbf{x}_t\} \tag{3}$$

The output of the LSTM decision layer is defined as the conditional probability of executing an early closing operation at the current time:

$$p_t = P(exit \mid \mathbf{X}_{t-k:t}) \tag{4}$$

This probability can be regarded as a probabilistic characterization of "unrealized profit drawdown risk." When $p_t$ is high, it indicates that the model believes that, if the position continues to be held, the current unrealized profit is more likely to experience a future drawdown. Conversely, when $p_t$ is low, it suggests that continuing to hold the position may still have a higher expected value than exiting immediately.

Therefore, the problem in this paper can be viewed as a sequential binary classification problem: given the current and historical market-state sequence $\mathbf{x}_{t-k:t}$, the model learns to determine whether the position should be actively exited before the reverse signal of the baseline strategy appears. In this way, the LSTM layer is transformed from a traditional "direction prediction model" into a "profit protection and risk control model."

## 4.4 Feature Engineering

To enable the LSTM decision layer to perceive both the market environment and the current trading state, this paper constructs an input feature system jointly composed of market features and position-context features, providing the informational basis required for the model to identify future unrealized profit drawdown risk.

### 4.4.1 Market Features

When selecting market features, this paper follows the approaches of Fischer and Krauss (2018), Huang et al. (2024), and AlKhatib et al. (2022), and jointly considers price trends, return-generating ability, candlestick patterns, volatility conditions, and trading activity. The detailed definitions of the variables are shown in Table 2.

Table 2: Market Features

| Category | Feature | Description |
|---|---|---|
| Trend | `wma30_rel`, `wma60_rel` | Deviation of the closing price from the moving averages |
| Trend | `wma_spread` | Deviation of the 30-day moving average from the 60-day moving average |
| Trend | `wma30_slope`, `wma60_slope` | First-order differences of the moving averages |
| Return | `log_ret`, `oc_ret` | Log return and open-close ratio |
| Pattern | `hl_range`, `shadow_imbalance` | Price range and upper-lower shadow imbalance |
| Volatility | `atr14_ret` | ATR(14) divided by the closing price |
| Trading Activity | `vol_z20`, `taker_buy_ratio` | Trading volume Z-score and active buy ratio |

### 4.4.2 Position Context Features

In addition to market features, the key innovation of this paper lies in the introduction of position-context features. Traditional technical indicators usually describe the market itself, but cannot represent the current trading state. However, for the early take-profit problem, the decision implications of the same market trend may differ across different stages of the holding period. For example, the drawdown immediately after opening a position and the drawdown after a large unrealized profit are not equivalent in terms of risk implications. Therefore, this paper represents the position state as model input. The detailed definitions of the variables are shown in Table 3.

Table 3: Position Context Features

| Feature | Description |
|---|---|
| `age` | Number of bars for which the current position has been held |
| `pnl_dir` | Unrealized profit or loss in the direction of the position |
| `mfe` | Maximum favorable excursion |
| `retrace` | Drawdown from the maximum favorable excursion |
| `dir` | Position direction, where "+1" denotes a long position and "-1" denotes a short position |

By introducing the above position-context information, the LSTM model no longer makes decisions solely based on market states, but is able to make decisions according to the dynamic profit and loss condition of the current trade, the holding duration, and other information. This allows the model to more closely approximate the risk-management logic of real trading.

## 4.5 Label Construction

To train the LSTM decision layer, this paper needs to construct a supervisory label indicating whether an early exit should be executed for each time point in the holding state. This paper adopts a value-based labeling method based on the trade-off between future returns and future drawdowns, so that the labels can more closely reflect unrealized profit drawdown risk.

For a trade generated by the baseline strategy, let its entry time be $t_0$ and the original exit time of the baseline strategy be $T$. At any holding time $t$, if the position is held until the exit point given by the baseline strategy, the future incremental return that may still be obtained relative to the current time is defined as:

$$future\_gain_t = pnl_T - pnl_t \tag{5}$$

where $pnl_t$ denotes the unrealized profit adjusted by the current position direction, and $pnl_T$ denotes the profit when the baseline strategy eventually exits.

At the same time, if the position continues to be held from the current time, the price may experience adverse fluctuations along the future path. This paper defines the future adverse drawdown as $future\_drawdown_t$, which is used to measure the maximum adverse price movement that the trade may encounter between the current time and the baseline exit time. Based on this, this paper constructs the value evaluation function:

$$value\_score_t = future\_gain_t - \lambda \cdot future\_drawdown_t \tag{6}$$

where $\lambda$ is the risk-penalty coefficient, used to reflect investors' aversion to risk. When $\lambda$ is larger, the label construction process places greater emphasis on drawdown risk; conversely, it places more emphasis on the future returns that may be obtained by continuing to hold the position.

This paper uses the value score as a proxy indicator for unrealized profit drawdown risk. If $value\_score_t < 0$, it indicates that the potential return from continuing to hold the position from the current time is insufficient to compensate for the adverse drawdown that may occur in the future. In this case, the sample should be labeled as an early exit:

$$y_t = \begin{cases} 1, & \text{if } value\_score_t < 0 \\ 0, & \text{if } value\_score_t \geq 0 \end{cases} \tag{7}$$

where $y_t = 1$ indicates that an early closing operation should be executed at the current time, while $y_t = 0$ indicates that the position should continue to be held. In this way, the model's learning objective is no longer to predict the price direction, but to identify whether the current position is exposed to relatively high future drawdown risk.

To prevent the model from being disturbed by noise in the early stage after opening a position, this paper sets a minimum holding-period constraint. A sample is included in the label construction process only when the holding time of the trade exceeds 90 minutes and the current unrealized profit exceeds 0.5%. This treatment ensures that the LSTM decision layer mainly serves the exit management after an unrealized profit has already emerged, rather than interfering with the initial entry logic of the baseline strategy.

## 4.6 LSTM Model

This paper adopts a Long Short-Term Memory (LSTM) network as the early-exit decision model. LSTM is suitable for capturing dynamic dependence in financial time series and can, to a certain extent, capture the nonlinear relationship between trend evolution and position-state changes.

The model input is a fixed-length historical feature matrix:

$$\mathbf{X}_{t-63:t} \in \mathbb{R}^{64\times 17} \tag{8}$$

where 64 denotes the length of the time window, and 17 denotes the feature dimension contained in each time step.

The LSTM model recursively encodes the input sequence and uses the hidden state of the last time step as a comprehensive representation of the current holding state. The hidden state is then mapped into a one-dimensional output through a fully connected layer and converted into a probability form through the Sigmoid function:

$$p_t = \sigma(\mathbf{w}\mathbf{h}_t + b) \tag{9}$$

where $\mathbf{h}_t$ denotes the hidden state of the LSTM at the current time, $\mathbf{w}$ and $b$ denote the weight and bias of the fully connected layer, respectively, and $\sigma(\cdot)$ denotes the Sigmoid activation function. The final output is interpreted as the probability of early exit at the current time, namely:

$$p_t = P(y_t = 1 \mid \mathbf{X}_{t-63:t}) \tag{10}$$

When $p_t$ is close to 1, it indicates that the model believes the future unrealized profit drawdown risk is relatively high and that the current position should be closed early. When $p_t$ is close to 0, it indicates that continuing to hold the position is more reasonable.

## 4.7 Decision Rule

In the backtesting stage, the LSTM decision layer operates as an additional exit module on top of the baseline strategy. The baseline strategy still generates trading directions and initial entry signals, while the LSTM model participates in decision-making only when an existing position is being held.

Specifically, when the system executes a WMA crossover rule based on the baseline strategy and establishes a position, for each new candlestick, the market features and position-context features are updated, and the feature sequence of the most recent 64 time steps is input into the LSTM model to obtain the early-exit probability $p_t$.

The exit rule in this paper can be expressed as:

$$Exit_t = \begin{cases} 1, & p_t > \theta \\ -1, & \text{opposite WMA crossover occurs} \\ 0, & \text{else} \end{cases} \tag{11}$$

where $\theta$ is the decision threshold for early exit, determined through grid search, and the results are shown in Table 4. When $p_t > \theta$, it indicates that the model judges the current position to face a relatively high future unrealized profit drawdown risk, and therefore the system will actively close the position before the reverse crossover signal of the baseline strategy appears. If the LSTM does not trigger an early exit, the trade continues to be managed according to the original moving-average reverse crossover rule.

Table 4: Backtesting Results under Different LSTM Decision Thresholds

| Threshold $\theta$ | Number of LSTM Exits | LSTM Exit Ratio (%) | Cumulative Return (%) | Win Rate (%) | Sharpe Ratio | Average Holding Bars | Maximum Drawdown |
|---|---|---|---|---|---|---|---|
| Baseline | 0 | 0.0 | +1.2807 | 36.7 | 2.252 | 48.9 | 0.2782 |
| 0.30 | 142 | 68.6 | +0.4161 | 68.6 | 3.723 | 11.2 | 0.1045 |
| 0.40 | 141 | 68.1 | +0.4511 | 68.1 | 3.842 | 11.6 | 0.1390 |
| 0.50 | 121 | 58.5 | +0.8222 | 64.3 | 3.960 | 18.6 | 0.1336 |
| 0.55 | 58 | 28.0 | +0.9526 | 51.2 | 2.602 | 34.2 | 0.1351 |
| $\geq 0.60$ | 0 | 0.0 | +1.2807 | 36.7 | 2.252 | 48.9 | 0.2782 |

Through the above decision rule, this paper forms a two-layer exit mechanism: the baseline strategy provides trend-reversal exits, while the LSTM layer provides risk-warning-based early exits. The former ensures that the strategy still follows trend-following trading logic, while the latter improves the responsiveness to unrealized profit drawdown risk during the holding period. This mechanism enables the strategy to no longer passively wait for the moving-average reverse crossover, but instead to lock in profits in advance when drawdown risk rises, thereby improving strategy performance.

# 5 Experimental Design

## 5.1 Dataset

To objectively evaluate the generalization ability of the LSTM-based early-closing decision framework in dynamic market environments, this study selects the cryptocurrency market as the empirical setting. The cryptocurrency market features 24/7 continuous trading, high volatility, and pronounced trend persistence, making it an ideal setting for testing whether the enhanced strategy can effectively improve the traditional trend-following model.

This study selects five mainstream cryptocurrency trading pairs as the experimental assets: BTC/USDT, ETH/USDT, SOL/USDT, XRP/USDT, and DOGE/USDT. These assets cover different market capitalizations, liquidity characteristics, and price fluctuation patterns, thereby allowing the robustness of the model across assets to be evaluated.

The raw market data used in the experiment range from January 1, 2022 to March 12, 2026. This interval fully covers multiple bull-bear cycles in the market, enabling the model to be tested across different market regimes, including trend continuation, range-bound fluctuation, and sharp drawdown. In this paper, 30-minute candlestick data are used for modeling. This time granularity ensures the reliability of trend signals while providing sufficient sequence depth for the LSTM model, so as to capture the dynamic evolution of micro risk structures during the holding period.

This study strictly follows the chronological order principle for predictive samples, and divides the dataset into the training set, validation set, and testing set, with proportions of approximately 70%, 15%, and 15%, respectively. The specific sample sizes and corresponding time intervals are shown in Table 5.

Table 5: Dataset Split

| Stage | Proportion | Sample Size | Time Interval | Research Function |
|---|---|---|---|---|
| Training Set | 70% | 29,250 | 2022/01/02–2025/01/06 | Model parameter learning and weight optimization |
| Validation Set | 15% | 6,268 | 2025/01/06–2025/08/05 | Hyperparameter tuning and overfitting prevention |
| Testing Set | 15% | 6,269 | 2025/08/05–2026/03/11 | Out-of-sample evaluation and robustness test |

## 5.2 Data Cleaning

This paper first conducts basic cleaning on the raw candlestick data, including uniformly parsing the time field into UTC time format, converting fields such as prices and trading volume into numerical types, and sorting the data in chronological order. Subsequently, anomaly auditing is performed on the raw sequence to identify missing candlesticks, zero-volume candlesticks, and extreme fluctuation samples, with corresponding flags constructed for each type of anomaly. Since the LSTM model adopts a fixed-length historical window for modeling, this paper further constructs a context-contamination flag using a rolling-window method, and removes samples whose historical windows contain bad data from the training dataset.

## 5.3 Comparison Methods

To examine whether the proposed Long Short-Term Memory (LSTM)-based early-closing decision framework can further improve performance on top of traditional trading strategies, this paper sets up several representative benchmark methods for comparative experiments. Specifically, the proposed LSTM early-closing strategy is compared with the WMA baseline strategy, the fixed take-profit strategy, and the trailing take-profit strategy, in order to evaluate its effectiveness in enhancing returns, controlling drawdowns, and improving risk-adjusted returns.

First, the WMA baseline strategy serves as the fundamental trend-following strategy in this paper, and is used to characterize the original strategy performance without introducing any additional early-closing mechanism. This method mainly relies on 30-day and 60-day weighted moving average signals for opening and closing positions, thereby providing a basic reference for evaluating whether subsequent exit-optimization methods are effective.

Second, the fixed take-profit strategy is a simple and commonly used rule-based exit method. It triggers position closing once the holding return reaches a fixed threshold, which is set to 10% in this paper. The advantage of this method lies in its clear rules and stable execution. However, its limitation is that it cannot adjust dynamically according to market conditions, which may lead to premature profit taking or failure to avoid profit drawdowns in a timely manner.

Third, the trailing take-profit strategy is used to represent a more dynamic traditional risk-control method. When the holding return reaches a preset threshold, which is set to 3% in this paper, the strategy enters a take-profit monitoring state. Thereafter, if the current holding return falls by more than a preset threshold from the highest unrealized profit level reached during the holding period, which is set to 10% in this paper, the position is closed. Compared with the fixed take-profit strategy, the trailing take-profit strategy can preserve the potential for further gains during trending markets to some extent while controlling the drawdown of unrealized profits. Therefore, it serves as an important benchmark for testing whether the proposed method has stronger dynamic risk-identification capability.

In terms of experimental design, this paper applies the above strategies uniformly to multiple mainstream cryptocurrency trading pairs, including BTC/USDT, ETH/USDT, SOL/USDT, XRP/USDT, and DOGE/USDT, and compares their performance across different assets. Through cross-asset experiments, the robustness of the proposed method can be further examined.

## 5.4 Evaluation Metrics

To comprehensively evaluate the performance of different strategies and models in the early-closing task, this paper constructs an evaluation metric system from five dimensions: profitability, risk-adjusted return, drawdown control, trading stability, and holding efficiency. Specifically, the metrics include cumulative return, Sharpe ratio, maximum drawdown, win rate, and average holding time.

First, cumulative return is used to measure the overall profitability of the strategy during the testing period. This metric reflects the growth of the initial capital after the complete backtesting period and is the most direct indicator for evaluating strategy return performance. It is calculated as follows:

$$Ret = \frac{V_T - V_0}{V_0} \tag{12}$$

where $V_0$ denotes the initial capital size, and $V_T$ denotes the capital size at the end of the backtesting period. A higher cumulative return indicates stronger overall profitability during the out-of-sample testing period.

Second, the Sharpe ratio is used to measure the excess return obtained by the strategy per unit of risk taken. Compared with a simple return metric, the Sharpe ratio further considers return volatility, and therefore can better reflect the risk-adjusted return performance of the strategy. It is calculated as follows:

$$SharpeR = \frac{R_p - R_f}{\sigma_p} \tag{13}$$

where $R_p$ denotes the strategy return, $R_f$ denotes the risk-free rate, and $\sigma_p$ denotes the standard deviation of strategy returns. In high-frequency or short-period cryptocurrency backtesting, since each holding period is relatively short, the risk-free rate can be approximately set to zero in this paper. A higher Sharpe ratio indicates that the strategy can obtain higher returns under the same level of risk.

Third, maximum drawdown is used to measure the maximum capital loss that the strategy may suffer during the backtesting period, and is an important indicator for evaluating the risk-control capability of the strategy. It is calculated as follows:

$$MDD = \max_{t \in [0,T]} \frac{V_t^{peak} - V_t}{V_t^{peak}} \tag{14}$$

where $V_t$ denotes the net asset value at time $t$, and $V_t^{peak}$ denotes the historical highest net asset value up to time $t$. A smaller maximum drawdown indicates stronger capital-protection capability under adverse market fluctuations. For an early-closing strategy, if the model can identify profit drawdown risk in a timely manner, it should reduce the maximum drawdown to some extent.

Fourth, the win rate is used to measure the stability of trading outcomes, and represents the proportion of profitable trades among all trades. It is calculated as follows:

$$WinRate = \frac{N_{win}}{N_{total}} \tag{15}$$

where $N_{win}$ denotes the number of profitable trades, and $N_{total}$ denotes the total number of trades. A higher win rate indicates that the strategy generates profitable trades more frequently. However, the win rate alone cannot represent the overall quality of a strategy, because a high-win-rate strategy may still produce poor overall returns due to a small number of large losses. Therefore, this paper analyzes it jointly with cumulative return, Sharpe ratio, and maximum drawdown.

Finally, average holding time is used to measure the average duration from opening to closing a position, reflecting the impact of different closing mechanisms on the holding period. It is calculated as follows:

$$AvgHT = \frac{1}{N}\sum_{i=1}^{N}\left(t_i^{exit} - t_i^{entry}\right) \tag{16}$$

where $t_i^{entry}$ and $t_i^{exit}$ denote the opening time and closing time of the $i$-th trade, respectively, and $N$ denotes the total number of trades. Average holding time can reflect whether a strategy tends to exit too early or hold positions for a long time. For the LSTM-based early-closing framework proposed in this paper, this metric helps determine whether the model can avoid subsequent drawdown risk in a timely manner while preserving the profit potential of trend-following trades.

# 6 Results

# 7 Discussion

# 8 Robustness Checks

# 9 Conclusion

# 10 Future Work

## References

Adegboye, A. and Kampouridis, M. (2021). Machine learning classification and regression models for predicting directional changes trend reversal in FX markets. *Expert Systems with Applications*, 173:114645.

AlKhatib, K., Khazaleh, H., Alkhazaleh, H. A., Al-Soud, A., and Abualigah, L. (2022). A new stock price forecasting method using active deep learning approach. *Journal of Open Innovation: Technology, Market, and Complexity*, 8(2):96.

Ayala, J., García-Torres, M., Vázquez Noguera, J. L., Gómez-Vela, F., and Divina, F. (2021). Technical analysis strategy optimization using a machine learning approach in stock market indices. *Knowledge-Based Systems*, 225:107119.

Borovkova, S. and Tsiamas, I. (2019). An ensemble of LSTM neural networks for high-frequency stock market classification. *Journal of Forecasting*, 38(6):600–619.

Chiarella, C., He, X.-Z., and Hommes, C. (2006). A dynamic analysis of moving average rules. *Journal of Economic Dynamics and Control*, 30(9–10):1729–1753.

Dai, B., Marshall, B. R., Nguyen, N. H., and Visaltanachoti, N. (2021). Risk reduction using trailing stop-loss rules. *International Review of Finance*, 21(4):1334–1352.

Dempster, M. A. H. and Jones, C. M. (2001). A real-time adaptive trading system using genetic programming. *Quantitative Finance*, 1(4):397–413.

Dempster, M. A. H. and Leemans, V. (2006). An automated FX trading system using adaptive reinforcement learning. *Expert Systems with Applications*, 30(3):543–552.

Ellis, C. A. and Parbery, S. A. (2005). Is smarter better? a comparison of adaptive, and simple moving average trading strategies. *Research in International Business and Finance*, 19(3):399–411.

Fischer, T. and Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669.

Gerlein, E. A., McGinnity, M., Belatreche, A., and Coleman, S. (2016). Evaluating machine learning classification for financial trading: An empirical approach. *Expert Systems with Applications*, 54:193–207.

Ghosh, P., Neufeld, A., and Sahoo, J. K. (2022). Forecasting directional movements of stock prices for intraday trading using LSTM and random forests. *Finance Research Letters*, 46(Part A):102280.

Glynn, P. W. and Iglehart, D. L. (1995). Trading securities using trailing stops. *Management Science*, 41(6):1096–1106.

Han, Y., Kim, J., and Enke, D. (2023). A machine learning trading system for the stock market based on N-period Min-Max labeling using XGBoost. *Expert Systems with Applications*, 211:118581.

Hansun, S. (2013). A new approach of moving average method in time series analysis. In *2013 Conference on New Media Studies (CoNMedia)*, pages 1–4. IEEE.

Huang, Y., Wan, X., Zhang, L., and Lu, X. (2024). A novel deep reinforcement learning framework with BiLSTM-Attention networks for algorithmic trading. *Expert Systems with Applications*, 240:122581.

Hunter, J. S. (1986). The exponentially weighted moving average. *Journal of Quality Technology*, 18(4):203–210.

Hwang, Y., Park, J., Lee, Y., and Lim, D.-Y. (2023). Stop-loss adjusted labels for machine learning-based trading of risky assets. *Finance Research Letters*, 58(Part A):104285.

Kim, T. and Kim, H. Y. (2019). Optimizing the pairs-trading strategy using deep reinforcement learning with trading and stop-loss boundaries. *Complexity*, 2019:3582516.

Leung, T. and Li, X. (2015). Optimal mean reversion trading with transaction costs and stop-loss exit. *International Journal of Theoretical and Applied Finance*, 18(3):1–31.

Leung, T. and Zhang, H. (2021). Optimal trading with a trailing stop. *Applied Mathematics and Optimization*, 83(2):669–698.

Lo, A. W. and Remorov, A. (2017). Stop-loss strategies with serial correlation, regime switching, and transaction costs. *Journal of Financial Markets*, 34:1–15.

Marshall, B. R., Nguyen, N. H., and Visaltanachoti, N. (2017). Time series momentum and moving average trading rules. *Quantitative Finance*, 17(3):405–421.

Mehtab, S., Sen, J., and Dutta, A. (2020). Stock price prediction using machine learning and LSTM-based deep learning models. arXiv preprint arXiv:2009.10819.

Nelson, D. M. Q., Pereira, A. C. M., and de Oliveira, R. A. (2017). Stock market's price movement prediction with LSTM neural networks. In *2017 International Joint Conference on Neural Networks (IJCNN)*, pages 1419–1426. IEEE.

Sebastião, H. and Godinho, P. (2021). Forecasting and trading cryptocurrencies with machine learning under changing market conditions. *Financial Innovation*, 7(1):1–30.

Shawky, T. (2023). Gold exchange traded funds (ETFs), take profit/stop loss prediction using machine learning. *Journal of Business and Management Studies*, 5(4):6–17.

Shen, S., Jiang, H., and Zhang, T. (2012). Stock market forecasting using machine learning algorithms. Technical report, Department of Electrical Engineering, Stanford University.

Sivri, M. S., Gultekin, A. B., Ustundag, A., Beyca, O. F., Ari, E., and Gurcan, O. F. (2023). An intelligent system for determination of stop-loss and take-profit limits: A dynamic decision learning approach. In Kahraman, C., Sari, I. U., Oztaysi, B., Cevik Onar, S., Cebi, S., and Tolga, A. ., editors, *Intelligent and Fuzzy Systems: Intelligence and Sustainable Future Proceedings of the INFUS 2023 Conference*, volume 759 of *Lecture Notes in Networks and Systems*, pages 617–624. Springer Nature.

Sun, S., Wang, R., and An, B. (2023). Reinforcement learning for quantitative trading. *ACM Transactions on Intelligent Systems and Technology*, 14(3):44:1–44:29.

Trippi, R. R. and DeSieno, D. (1992). Trading equity index futures with a neural network. *Journal of Portfolio Management*, 19(1):27–33.

Tsantekidis, A., Passalis, N., Toufa, A.-S., Saitas-Zarkias, K., Chairistanidis, S., and Tefas, A. (2021). Price trailing for financial trading using deep reinforcement learning. *IEEE Transactions on Neural Networks and Learning Systems*, 32(7):2837–2846.

Vezeris, D., Kyrgos, T., and Schinas, C. (2018). Take profit and stop loss trading strategies comparison in combination with an MACD trading system. *Journal of Risk and Financial Management*, 11(3):56.

Wang, Z., Huang, B., Tu, S., Zhang, K., and Xu, L. (2021). DeepTrader: A deep reinforcement learning approach for risk-return balanced portfolio management with market conditions embedding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 643–650.

Zhong, X. and Enke, D. (2019). Predicting the daily return direction of the stock market using hybrid machine learning algorithms. *Financial Innovation*, 5(1):1–20.

# A Additional Tables

Table 6: Variable Definitions

| Variable | Definition |
|---|---|
| $Flow_t^{5d}$ | Five-day change in stablecoin market capitalization |
| $\Delta \ln Flat_t$ | Growth rate of fiat-reserve-backed stablecoin market capitalization |
| $\Delta \ln Coll_t$ | Growth rate of collateralized stablecoin market capitalization |
| $\Delta y_t$ | Change in Treasury yield |
| $X_t$ | Vector of control variables |

# B Additional Figures

Additional figures can be placed here.